Changelog

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2026-10-15 - 2.0.0: a native window and a console command

Added

- A native Windows window (C#, Avalonia): `Start.exe` opens `App\AudionOfficeOcrAi.exe`. The window carries its own .NET and its own PowerShell (`App\AudionPs.exe`), so nothing is installed.
- `SETUP` in the header, before `SETTINGS`: the installation of what the machine needs for local OCR; while the machine has no Tesseract, the window opens on it at start. Five sections in one band under the header: `LOCAL OCR`, `API OCR`, `OCR CHECK`, `TABLES`, `TOOLS`; the quick conversion of office files into Markdown is `TOOLS` -> `To Markdown`. The engines of `LOCAL OCR` and the providers of `API OCR` are switches over the form; a form stands in named blocks with a coloured stripe, rarely needed settings under `ADDITIONAL`, risky choices in red. The marks of Yandex, Mistral, Gemini and ChatGPT wear their brand colours.
- `ocr_cli.cmd`: every operation of the window runs from a console with the same settings - `describe`, `run --operation ID [--source] [--target] [--set NAME=VALUE]`, `options` (`--refresh` asks the provider), `key add|remove`, `hardware`, `version`. Exit codes: 0 done, 1 failed, 130 stopped, 2 a wrong call - refused before anything runs. The window writes the line it runs into its terminal.

- The header names what the machine has for OCR: the recommendation for its video card, Tesseract and its languages, the layout service of Surya, the providers that have a key - a count of key files, never a key.
- The layout service of Surya is started and stopped from the line under the `SURYA` switch.
- A key of a provider is added and set aside from its field: it is pasted into a dialog, goes to the console command by its standard input and is shown nowhere; a key set aside moves to `config\deleted_keys\`.
- The lists of models open under their field at once, from what the program keeps; `REFRESH THE LIST` asks the provider.
- The working area: `SOURCE` (a folder or one file, dropped on the window or picked) and `TARGET` with a history of paths and pins; `LIST` names what lies in the source.

Changed

- A run with a `SOURCE` outside the program asks first when `input\` or `output\` hold anything, and names how many items each holds: the source is copied into `input\`, and both are cleared before it. A `SOURCE` inside `output\` and a folder inside `input\` are refused before anything runs.
- One clearing for the working area: the contents of `SOURCE`, `TARGET` or both into the Recycle Bin, and the DocumentModel artifacts, each with its question; the root of a drive, the program itself and links are refused.
- `STOP` asks the operation to stop at its next safe point and ends it at once on a second press; closing the window while an operation goes asks first and waits for the stop.
- The terminal shows the colours of the operations and one live line of what a run is doing.
- The output of the programs an operation runs reaches the terminal as text whatever they write in: UTF-8, the OEM and the ANSI code pages, and now UTF-16 too; the byte order mark of UTF-8 no longer stands at the start of a line. The lines of the Surya layout service script, Russian messages of PowerShell among them, are no longer question marks.
- `Layout.Service.ps1 -Disable` stops only the layout service of its own copy of the program.
- The environment is built anew from the current packages of `install\requirements_full.in`; with google-genai 2.31 the request to Gemini names the type of a page image `mime_type`.
- The install scripts take `App\AudionPs.exe` first. `install\verify_portable_env.cmd` checks the console command, Tesseract, `Start.exe` and the self-check of the window. `builder_main.cmd` gains `START LAUNCHER` and `WINDOW`; the steps of the OCR engines keep their numbers.

Removed

- The window on NiceGUI in a pywebview frame, the console menus (`launcher_gui.cmd`, `launcher_project.cmd`, `launcher_project_ru.cmd`, `launcher_tools.cmd`), fzf and the portable PowerShell. The licences are collected by Audion Build Licenses and the release is made by Audion Build Publisher, so `install\make_release_archive.cmd` and `system_core\license\` are gone too.
- NiceGUI and pywebview leave `install\requirements_full.in`; the environment built next holds only what the engine needs.
- What nothing in the program used: the HTML terminal of the former window, the decoders and helpers only it called, a stub of Yandex batch OCR, the second example folder of Yandex keys under `system_core\ocr_brick\config`, the unused `requirements.txt` of OCR Brick and `ocr_named_entities_ru.md` (the hints are read from `ocr_named_entities_ru.yaml`).
- `DEV Markdown PDF` with its engine and its Chromium (`runtime\.playwright`, step `[05]` of `builder_main.cmd`, the package `playwright`): paired dark and light PDFs of Markdown are made by Audion Build PDF Render (`render_markdown.cmd`). The PDF of the program stays Word-based (`Build PDF`).

2026-07-19 - Resilient bootstrap downloads

Changed

- Restored the Workbench path/deletion helper chain so an unpinned external Source selected from history reaches the operation snapshot exactly like the native picker route.
- FZF and portable PowerShell installers now resolve and download deterministic release assets without consuming GitHub REST quota in the normal path, while preserving API metadata as a fallback when direct asset conventions change.
- Real-ESRGAN remains on its metadata-driven release resolver because its Windows portable asset does not follow the same simple latest-version contract as the bootstrap utilities.

2026-07-17 - Canonical Workbench and shared CLI/GUI services

Changed

- Transplanted the byte-identical canonical Workbench with the shared `Source`, `Add file...`, `Target`, `Reset`, `Delete`, and `List` vocabulary.

- Added protected single-file/folder pickers, process-tree cleanup, local-only GUI host enforcement and the sanitized native-window shutdown lifecycle.
- Routed GUI Workbench paths as per-operation Source/Target snapshots while preserving managed `input`/`output` roots for the backend mirror contract.
- Added `system_core/cli_operation.py`; both CLI/FZF launchers now execute the same manifest/service operations as NiceGUI for OCR Brick, Workbench resolvers and office builders.
- Replaced obsolete CLI-only OpenAI/Gemini OCR copies with the current Tesseract, Surya, Yandex, Mistral, xAI, Gemini and OpenAI OCR Brick routes.
- Hardened cleanup and release staging: `.git` is preserved, while user documents, reports, caches, local route history and Yandex credential metadata are excluded from portable archives.

Verified

- Added single-file mirror/Target synchronization tests, CLI/GUI dispatch parity checks and an offline Tesseract-to-DocumentModel/DOCX/JSON/Markdown integration smoke.
- Removed the unused legacy `extract_ai` service and generic model selector after confirming that no manifest, GUI, CLI or test referenced them.

2026-07-15 - Provider-neutral DocumentModel and office exports

Added

- Added the lossless `<name>.document` package as the canonical OCR layer, retaining exact sources/page rasters, hashes, all OCR candidates, coordinates, structured regions, table spans, confidence, verification and manual-correction slots.
- Added local DOCX, Searchable PDF, XLSX, ODT, Markdown, HTML, OCR JSON and full ZIP exporters. They do not rerun paid OCR and do not require LibreOffice.
- Added independent checkbox-chip output selection backed by the canonical DocumentModel.
- Added automatic local Tesseract coordinate generation when a cloud OCR result is exported as Searchable PDF.
- Capped adaptive DOCX page size at A3.
- Added content-aware column balancing for DOCX and XLSX: compact numeric/year columns yield space to names, addresses and other long text.
- Added Mistral second-pass GUI controls: none, Tesseract, Yandex, or Yandex plus Tesseract, with suspicious/all-page scope and optional text fusion.

Changed

- Output formats now use a balanced 4-by-2 checkbox-chip grid: four office formats and four developer formats. Full archive ZIP remains supported by the backend but is no longer exposed in the primary OCR GUI.
- Removed the duplicated `Output formats` and `Scan cleanup` field labels while retaining their shaded explanatory hint blocks; the section headings `Deliverables` and `Preprocessing` remain the only upper headings.
- Restored visual separation for deliverable chips by placing the complete 4-by-2 checkbox grid on the same subdued dark container treatment used by segmented button groups.
- Expanded the two OCR output-contract buttons into equal full-row columns for better visual balance.
- Added conservative Mistral/Yandex fusion. Table cells use verified physical geometry, provider-native coordinates or unambiguous native Tesseract word anchors; unverified grids never authorize replacement. Numeric disagreements and ambiguous coordinates remain review items.
- Added a physical ruled-table model: long raster lines define rows/columns, the OCR numbering row verifies the column count, header merges are recovered from missing separators, and all provider text is assigned inside exact cells.
- Added Yandex candidates, provider verification and per-item fusion decisions to DocumentModel 1.1 and the XLSX audit sheet.
- Fixed the paid OCR root filter so Mistral OCR 4 is actually visible in the GUI.
- Kept the OCR engine selector in one horizontal row with a single `Engine` / `Движок` heading. Engine buttons now divide the full section width evenly: two local engines use two equal columns, and five API engines use five equal columns.
- Promoted the shared `Deliverables` / `Готовые файлы` block directly below the engine selector for both local and paid OCR. Preset buttons were removed; formats are independent checkboxes and DOCX is the working default. The internal DocumentModel is always preserved.
- Added a confirmed maintenance action `Clear DocumentModel artifacts` / `Очистить DocumentModel`. It removes regenerable `*.document`, `*.document.json` and `*.verification.json` artifacts from managed output/report/workspace while preserving DOCX, XLSX, Markdown, PDF, images and sources.
- Removed the obsolete `Build office output` card from the main GUI. The old Markdown compilers remain available under `Project tools -> Rebuild legacy Markdown` for compatibility with archived Markdown workflows.
- Reworked OCR deliverables into a balanced 4-by-2 grid of rounded rectangular checkbox chips: four equal amber office columns and four equal sea-green dev/machine columns. The chip outline

is intentionally very subtle, while the square checkbox keeps a clearer but slightly muted outline; narrow layouts fall back to two columns.

- Flattened the `Tables` window into three expanded operation blocks with their own parameters and Run buttons; coordinate extraction, Markdown table inspection and XLSX build no longer require child navigation.
- Refreshed the Russian and English documentation stack around the current DocumentModel-first workflow, direct 4-by-2 deliverables grid, conservative table geometry, maintenance semantics and legacy-only Markdown rebuild path.

Clarified

- "100% completeness" means no loss of source data; it is not a claim of 100% OCR accuracy.

2026-07-06 - DEV Markdown PDF and GUI polish

Changed

- Changed the DEV Markdown PDF default output to `docs/PDF` in the GUI, service fallback and standalone renderer.
- Reworked DEV Markdown PDF source mode into compact top-level buttons: `Источник`, `input/output`, `Документация`, `PDF-пары`, `Устаревшие PDF`.
- Kept `PDF рядом с MD` as an explicit alternative: every folder with Markdown gets one shared `PDF` subfolder next to those Markdown files.
- Clarified the other DEV Markdown PDF output modes: `Прямо рядом с MD`, `Общая docs/PDF`, and `Зеркало в Назначение` through the current Workbench Target.
- Replaced the old external dev-PDF folder wording in the GUI-facing documentation model.
- Compact hardware recommendation badge is now shown only in `Инструменты проекта` and `Локальный OCR`.
- Kept canonical tooltip styling and detailed control tooltips while skipping noisy per-checkbox tooltips for document-kind chips.

2026-07-01 - OCR Brick integration and UI cleanup

Added

- Added root-level OCR windows:
  - `Офисный текст без OCR`

    - `Локальный OCR`
    - `Платный API OCR`
    - `Тест качества OCR`
    - `Проверка реквизитов`
    - `Таблицы`
- Added OCR Brick pipeline under `system_core/ocr_brick` with rasterize, preprocess, cache, OCR engine dispatch and Markdown/JSON output.
- Added CPU preprocessing fallback with PIL/NumPy: JPEG denoise, autocontrast, contrast boost, unsharp mask, adaptive binarization, deskew and vertical-line removal.
- Added Real-ESRGAN ncnn-vulkan portable installer and detection from `tools/realesrgan-ncnn-vulkan`.
- Added engine-aware scan cleanup profile `Очистка скана`:
    - `Авто` uses tuned defaults for the selected OCR engine.
    - `Без очистки` sends the page raw.
    - `Тяжёлый скан` uses stronger denoise, contrast and sharpening.
    - `Цифры` boosts legal/financial identifiers.
    - `Ручной` exposes the old detailed preprocessing switches in Advanced.
- Added local OCR engines:
    - Tesseract as the recommended local baseline.
    - Surya as optional slow/local OCR for cases where API cost matters more than time.
- Added paid/API OCR engines:
    - Yandex Vision OCR with selectable mode/model and OCR-safe model cache.
    - xAI vision OCR with model list refresh, curated pinned models and API smoke test.
    - Gemini and OpenAI vision OCR through the same OCR Brick vision contract.
- Added Gemini API cost/scheduling controls:
    - `Gemini tier` selector with `Standard` and `Flex` for regular OCR and one-page OCR quality tests.
    - `Gemini Batch OCR` project tool to prepare local JSONL manifests and optionally submit asynchronous Batch API jobs.
    - `Забрать Gemini Batch` project tool to poll/download finished Batch OCR results into Markdown.
- Added `2-pass Tesseract` in paid OCR:

    - For Yandex it writes a local sidecar with Tesseract text/boxes.
    - For xAI/Gemini/OpenAI it can inject Tesseract text as a fallible prompt hint.
- Added one-page OCR quality benchmark for raw/clean comparisons before large OCR runs.
- Added numeric/legal requisites checker `Проверка реквизитов` with Tesseract, Surya or vision-page locator modes and xAI verification.
- Added global tooltips for command buttons, fields, model selectors, service buttons and API-key controls.
- Added API key add/delete UI with warnings and provider-specific key selectors.
- Added pinned model storage in `config/gui_model_pins.json`.
- Added hardware recommendation badge for local OCR and project tools. It detects GPU class and recommends Vulkan, llama.cpp CUDA or PyTorch CUDA install paths.
- Added project-tools mode switcher: `Установка` / `Проверка`.

Changed

- Split the previous mixed OCR Brick window into separate local and paid OCR root windows.
- Reworked local/paid OCR dialogs into logical visual blocks: engine/model/access, preprocessing, local 2-pass, prompt/parameters, postprocessing, output and advanced.
- Replaced the visible preprocessing control field with a single engine-aware profile selector; detailed knobs are hidden until `Ручной` is selected.
- Reworked model selectors into wide dropdowns with pin/unpin/update controls and curated defaults.
- Shortened noisy operation labels and tightened the main root-menu row spacing.
- Moved coordinate-table extraction to root as `Таблицы`.
- Moved Tesseract install into `Инструменты проекта -> Установка` and marked it as recommended.
- Moved project checks, model refreshes and OCR smoke tests into `Инструменты проекта -> Проверка`.
- Updated install/build scripts for the current portable runtime layout and optional OCR engines.
- Updated DEV Markdown PDF output to support project `docs/PDF`, beside-source output and external target trees.
- Updated vision OCR prompt handling to reduce decorative Markdown and preserve source text more strictly.

- Updated Gemini OCR calls to pass `service_tier=flex` when Flex is selected and use longer timeouts for Flex queueing.
- Replaced stale user guides with current root-window documentation and moved implementation history into this changelog.
- Added `docs/GEMINI_BATCH_FLEX.md` with Gemini Standard/Flex/Batch usage, GUI mapping, artifact paths and the verified `Вопрос 239.pdf` dry-run.
- Updated GitHub README files to describe OCR Brick root windows, current local/API OCR engines, Gemini Batch/Flex and one-page quality/requisites workflow instead of the legacy Workbench-first flow.
- Added install documentation for optional OCR acceleration entries `[11]`-`[17]`, including practical RTX 50xx/40xx/Vulkan/CPU choices and the distinction between Real-ESRGAN, Surya, llama.cpp and heavy PyTorch CUDA benchmark payloads.

Removed

- Removed Paddle OCR from UI, install paths and project cleanup expectations.
- Hid/removed legacy Python OCR Workbench from the visible root navigation.
- Removed stale `config/keys` initialization from folder setup.
- Removed legacy mixed root OCR actions in favor of the new local/paid OCR windows.

Verified

- NiceGUI smoke test passes after UI changes.
- Python compile checks pass for updated UI/service files.
- Yandex OCR smoke passed with configured key/folder.
- xAI model-list and OCR smoke passed.
- Gemini Batch dry-run prepared 3 JSONL requests for `Вопрос 239.pdf` pages 6-8 without remote submission.
- Tesseract OCR Brick runs were smoke-tested on `10_Самарская_Тольятти.pdf` and `1_Псковская_Псковский.pdf`.
- Numeric check matched key Pskov identifiers with local locator plus xAI verification.

Cleanup

- Legacy implementation plans, handoff briefs and generated benchmark/log/cache artifacts were removed from the working tree after this changelog captured the relevant decisions.

- Source test package and smoke-run workspaces were removed from the portable project cleanup set.